Supplemental Material for "An Agent-Based Model for the Social Collapse of Calhoun's Universe 25 Experiment"

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This Supplemental Material gives the material needed to check the main text without the extended version: the full table of patterns and the numbers taken from Calhoun's report (Sec. S1), the production ensembles in full (Sec. S2), the null models with every criterion and with the preregistered evaluator (Sec. S3), the confirmatory family with both evaluators (Sec. S4), the transplant experiments (Sec. S5), the robustness variants and the lattice sizes (Sec. S6), the list of ensembles with their seeds (Sec. S7), the deviations from the preregistered plan (Sec. S8), a condensed description of the model following the ODD protocol (Sec. S9) and further details of the model (Sec. S10). For the calibration, the mean-field analysis, the phase planes, the global sensitivity analysis in full and the discussion of the comparison with Universe 25 we refer to the extended version in the repository (`https://github.com/jfbonder/universe25-abm`), which also contains the full ODD description, the code and the data. Candidates: C0 (R1 + R2 + AV), C1′ (R1 + R2 + R7), C1 (C1′ + AV) and C2 (C1 with the retreat driven by maternal socialization).

S1. PATTERNS AND NUMBERS TAKEN FROM CALHOUN'S REPORT

Table S1 lists the seventeen patterns and four anti-patterns with their full operational criteria, and Table S2 the numbers taken from the original report [1] and their conversion to weeks. Every criterion is evaluated per run and returns PASS, MARGINAL, FAIL or NOT APPLICABLE; an ensemble is summarized by the fraction of PASS runs with a 95% Wilson interval, and an essential (E) pattern is reproduced if that fraction is at least 0.8, a desirable (D) pattern if it is at least 0.5 [2]. The primary evaluator replaces t_{pk} by the start of the plateau t'_{pk} (the first week with $N \geq 0.98\, N_{\max}$) in every criterion and, in O5, t_{B0} by the week t_{B99} at which 99% of the births have occurred; f_0 is measured at t'_{pk} and the crowded class at $m > 8$.

TABLE S1: Qualitative patterns (O) and anti-patterns (A). E = essential, D = desirable, X = exploratory. †O13p is not preregistered (defined after calibration) and enters the primary outcome O17p = O17 ∧ O13p. O17p is strict: a not-applicable essential pattern counts as a failure (except O1–O3 when $N_0 > 20$, which are excluded by design), and an evaluation error is never a pass. O17 alone counts not-applicable patterns as passes, so it is not used as an outcome. ‡Consistency check of the design: the pattern follows from the rules by construction (O12 from R1 and $p_{\mathrm{conc}} \propto s_f s_m$; O13p from R7 with $c_R = 2$; O8 and A4 from $\mu_s = 0$). §Not preregistered: the preregistered form also passes in null models (Table S5); in the primary form O4 and O10 are still generic. $K_\gamma = m_c|G|$ is the population at which every cell would sit at the damage threshold; it is not a physical capacity. Page numbers refer to [1].

ID		Observation in Universe 25 (page)	Operational criterion
O1	E	104 days of "considerable social turmoil" before the first litters (p. 82)	founder scenario only: $t_{\mathrm{lag}}/t_{\mathrm{pk}} \in [0.10, 0.35]$ (Calhoun 0.19)
O2	E	Exponential growth, doubling time "about 55 days" (p. 83)	$R^2 \geq 0.90$ for $\log N$ in the growth window
O3	E	Doubling time rises to ≈145 days from $N = 620$ (pp. 83–84)	$r_B/r_C \geq 2$, phase C lasting $\geq 0.1\, t_{\mathrm{pk}}$
O4	E§	Peak of 2200 versus shelter for 3840, food for 9500 and water for 6144; 20% of nest boxes empty at the peak (p. 82)	$N_{\mathrm{pk}}/K_\gamma \leq 0.8$ and empty fraction of cells at the peak $f_0 \geq 0.10$

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ID		Observation (cont.)	Criterion
O5	E	Growth "abruptly ceased" on day 560; last young surviving weaning born around day 600 (p. 85)	$N(t_{B0})/N_{\mathrm{pk}} \geq 0.7$ and $(t_{B0} - t_{\mathrm{pk}})/t_{\mathrm{pk}} \leq 0.5$
O6	E	No recovery; the demise "contradicts prior knowledge" that remnant groups reinitiate growth (p. 85)	no rebound of $\geq 0.1N_{\mathrm{pk}}$, including slow regrowth still rising at censoring; birth-rate hysteresis $A_H \leq 0.1$
O7	E	Total demise; last male projected dead on day 1780 (p. 85)	run: extinct by T; ensemble: $P_{\mathrm{ext}}(T) \geq 0.9$ (Kaplan–Meier)
O8	D‡	Decline through senescence; mean age of survivors 776 days (p. 85)	$(T_{\mathrm{ext}} - t_{\mathrm{pk}})/t_{\mathrm{pk}} \geq 1$ (Calhoun ≥ 1.84 observed, 2.2 projected)
O8b	D	Slow initial decline: $N = 2056$ on day 736 (p. 83)	$N(1.31\,t_{\mathrm{pk}})/N_{\mathrm{pk}} \geq 0.8$ (Calhoun 0.93)
O9	D	27 survivors (23 F, 4 M), all older than 987 days (p. 83)	no young in the remnant; female fraction > 0.5
O10	E§	"death of societal organization by the end of Phase C" (p. 85)	adult mean social state falls below 0.5 before the peak and stays below it at the peak: $t(\bar{s}_{\mathrm{ad}} < 0.5) < t_{\mathrm{pk}}$, $\bar{s}_{\mathrm{ad}}(t_{\mathrm{pk}}) < 0.5$
O10b	D§	Phase B: 14 functional social groups during growth (days 104–315; pp. 83–84)	adults socially competent during growth: $t(\bar{s}_{\mathrm{ad}} < 0.5) \geq 0.5\,t_{\mathrm{pk}}$
O11	E§	From mid-phase C young "prematurely rejected by their mothers"; complex behaviours fail to mature (p. 85)	social state at maturity decreasing across cohorts, and < 0.2 for the cohorts born in the second half of natality; not applicable with fewer than 30 such animals (counts as failure in O17p)
O11b	D	Only 18% of 148 late-phase-C females had ever conceived (p. 85)	fraction ever conceived ≤ 0.3
O12	E‡	Small groups moved to uncrowded universes in mid-phase D failed to reproduce, even with partners raised without crowding (pp. 85–86)	transplant in silico: $P_T(D) \leq 0.1$, $P_T(B) \geq 0.5$
O13	E	Withdrawn males pooled at the centre, wounded by other withdrawn males; "beautiful ones" unwounded, never courting or fighting (pp. 84–85, 88)	instantaneous coexistence of crowded ($m > 8$) and isolated ($m \leq 2$) deteriorated classes, each $\geq 10\%$ of adults
O13p	E†‡	As O13, with a class that persists	isolated class persistent (≥ 4 consecutive weeks)
O13b	D	Beautiful ones are a late, unsocialized cohort, "the last half of the population born" (p. 85)	isolated class born later than crowded class; $\geq 50\%$ born after $0.8\,t_{\mathrm{pk}}$ (post hoc variant: after the median birth time)
O14	D	"Behavioural sink" on some hoppers (p. 81); births from 13 to 111 per brood group (pp. 83–84)	dispersion index ≥ 2 at the peak; birth Gini above the $\alpha = 0$ control
O15	D	Conception declined, foetal resorption and pre-weaning loss increased; no surviving young after day 600 (p. 85)	infant mortality increasing, ≥ 0.9 after t_{B0}
O15b	D	Conceptions continued until about day 920 (p. 85)	$(t_{C0} - t_{B0})/t_{\mathrm{pk}} \geq 0.1$ (Calhoun 0.57)
O16	X	Nursing females aggressive towards own young; withdrawn males attack each other (pp. 84–85)	only with aggression rules
O17	–	Robustness criterion, not an observation. Motivated by the single realization: "it was historical" (p. 88)	run is Calhoun-like: all applicable E patterns pass, no anti-pattern; ensemble fraction ≥ 0.8. Reported only through the strict primary outcome O17p
A1		Logistic saturation at capacity	plateau within 10% of N_{pk} for ≥ 100 weeks
A2		Cycles with recovery of births	births resume to $\geq 10\%$ of maximum, or ≥ 2 rebounds
A3		Demographic collapse before social collapse	per-capita fecundity halves only after t_{pk}
A4	‡	Adult death from deterioration as main cause	social share of deaths ≥ 0.5

S2. PRODUCTION ENSEMBLES

Table S3 gives, for the candidates and controls, the fraction of runs that pass each pattern with the primary evaluator, the transplants, the anti-patterns, the magnitudes and the timings of the boom. C1, C1′, C0 and the $\alpha = 0$ control come from one reference ensemble ($n = 200$, $T = 600$); C2 from its production ensemble ($n = 200$, $T = 700$); the short-lived control ($\alpha = 1.4$, $a_{\max} = 60$) and the long-lived control ($\alpha = 1.4$, $a_{\max} = 120$), the core model with the four rules switched off, from ensembles with $n = 100$ and $T = 1000$. In the 200 runs of C1 (medians with interquartile ranges) the population first comes within 2% of its maximum at week 34 [32, 36] and leaves that band at week 86 [85, 86]; the maximum itself is reached at week 55 [50, 60] and coincides with the last surviving birth in 69% of the runs;

TABLE S2. Numbers reported in [1] (days after colonization, DAC) and their conversion to weeks. The model has no weaning period: "last young surviving weaning" is compared with the end of surviving births in the model (t_{B0}, pups that survive birth), which is an earlier event in the life of a pup.

Event	DAC	weeks	page
4 pairs of 48-day-old BALB/c mice introduced (9 July 1968)	0	0	82
First litters (end of phase A)	104	14.9	82
End of phase B: 620 weaned (150 adults in 14 groups, 470 juveniles)	315	45	83–84
Peak, $N = 2200$	560	80	83, 85
Last young surviving weaning	∼600	85.7	83, 85
$N = 2056$	736	105	83
Last conception	∼920	131	85
122 survivors (22 M, 100 F)	1444	206	85
27 survivors (23 F, 4 M), all older than 987 days	1588	227	83
Projected death of last male	1780	254	85

the 99th percentile of the births falls at $1.17\,t'_{\mathrm{pk}}$ [1.12, 1.24], and the decline ratio is $(T_{\mathrm{ext}} - t'_{\mathrm{pk}})/t'_{\mathrm{pk}} = 3.97$ [3.72, 4.29] against 2.07 [1.85, 2.33] with $\arg\max N$. One C1′ run in 200 was still alive at $T = 600$ (it fails O7 and is censored in the survival analysis); no C1 run escaped. The Kaplan–Meier medians of the extinction time are 169 [167, 171] weeks for C1, 166 [165, 167] for C0, 182 [180, 185] for C1′, 218 [217, 222] for C2 and 186 [181, 190] for the long-lived control; the short-lived control is a mixture (47 of 100 colonies died by $T = 1000$, 44% [35, 54] in the first collapse, the others in a quasi-stationary oscillation with a tail hazard of 0.7×10^{-4} per week).

S3. NULL MODELS, EVALUATOR OPTIONS AND CLASS THRESHOLDS

Table S4 gives the discriminating power of every criterion, essential and desirable, in the reference ensemble, the four null models and the short-lived control; Table S5 repeats the redefined criteria with the preregistered evaluator, and Table S6 gives O17p for the main arms with each option of the primary evaluator changed separately. The peak-time definition matters only through O5: with t'_{pk} and the last sustained birth, O5 fails in half of the C1 runs (O17p 0.48), and taking the end of effective natality at 99% of the births removes this artefact (0.995); with $\arg\max N$ the result is insensitive to the birth reference (1.00). The fixed crowded-class threshold ($m > 8$) changes nothing at the C1 values, where $m_c = 8$; it lowers C2 from 0.78 to 0.70 together with the primary O4. The primary O10 and O11 remove the vacuous passes of the preregistered criteria: the preregistered O10 passed in every null model, and the preregistered O11 in 1.00 of the $\gamma = 0$ runs and in 0.09–0.21 of the other null models.

a. O11 at the end of the plasticity window. Re-scoring the same runs with the late cohorts measured at the age $a_w = 12$ instead of six weeks (same thresholds), C1 and C1′ barely change (O11 0.995 and 0.935, O17p 0.990 and 0.815), but O11 then also passes in 0.81 [0.75, 0.86] of the long-lived runs and in 0.755 [0.69, 0.81] of the runs without rules, so it no longer discriminates against those two null models; against $-\mathrm{R2}^*$ it passes in 0.21 (O17p 0.205), and without crowding damage it stays NA. The late cohorts of the long-lived and rule-free null models even end lower than those of C1 (median $\bar{s}$ 0.074 and 0.055 against 0.130). Pups born above the deterioration threshold ($s_{\mathrm{base}} = 0.25$ or 0.5) give O11 of 0.08 and 0.00 at six weeks, but 0.975 and 0.785 at the age a_w.

b. Threshold of the deteriorated classes. O13 requires both classes to hold at least $\theta = 10\%$ of the adults, and O13p the same of the persistent isolated class. Re-scoring the stored class fractions for other θ, C1 holds O17p = 0.99 up to $\theta = 0.125$ and 0.96 at 0.15 and collapses at 0.175; C1′ holds 0.83 up to 0.125 and falls to 0.33 at 0.15 (0.31 in the −AV arm of the factorial, $n = 120$); C2 falls from 0.70 to 0.01 at 0.125. On the grid of class thresholds $\theta \in \{0.05, \ldots, 0.20\}$ and ensemble thresholds $\tau \in \{0.5, \ldots, 0.9\}$, R1, R2 and R7 are necessary everywhere; AV is necessary exactly when $\tau \geq 0.8$ at $\theta \leq 0.125$ (0.79 lies below 0.8, with an interval [0.71, 0.85] that contains it) or when $\theta \geq 0.15$. In the hypercubes, the Calhoun-like fraction is 5.5%, 2.3% and 0.5% at $\theta = 0.05$, 0.10 and 0.15. The conclusions about refuges are therefore conditional on the 10% threshold, through the capacity bound $\Phi_{\mathrm{BO}}^{\mathrm{pers}} \leq f_R c_R |G|/N_{\mathrm{pk}}$.

S4. CONFIRMATORY FAMILY WITH BOTH EVALUATORS

Table S7 gives the preregistered family with the primary evaluator and with the preregistered evaluator on the same runs (Holm correction over seven tests within each set). Q1: the transplanted C1 animals are 37–64 weeks

TABLE S3. Production ensembles in full: essential patterns (fraction of applicable runs that pass; O17 and O17p over all runs), transplants (founding transplants over trials, 60 replicates per arm; O12b pools both sexes with healthy partners; phase D at $1.75\,t'_{\mathrm{pk}}$ of each colony and at week 95 in every arm), anti-patterns, magnitudes (ensemble means, except the Kaplan–Meier median and the spread of the extinction time, the interquartile range over the median of the Kaplan–Meier estimate), desirable patterns (fraction of passing runs), timings (medians over runs) and further magnitudes. "PASS or MARGINAL" for O13b counts the runs in which the evaluator finds the isolated class later-born at least marginally. The 95% Wilson half-width is at most 0.07 for $n = 200$ and 0.10 for $n = 100$.

ID	Pattern	C1	C1′	C0	α=0	C2	short-lived	long-lived
O4	peak below capacity, free space	1.00	1.00	1.00	0.00	0.90	0.87	0.08
O5	natality ends at high N	0.99	0.85	1.00	1.00	0.81	0.25	0.90
O6	no rebound	1.00	1.00	1.00	1.00	0.98	0.45	0.97
O7	extinction	1.00	0.99	1.00	1.00	0.98	0.47	0.98
O10	adult deterioration before peak	1.00	1.00	1.00	1.00	1.00	1.00	1.00
O11	late cohorts fail	1.00	0.97	1.00	1.00	0.99	0.00	0.00
O13	two deteriorated classes	1.00	1.00	0.92	1.00	1.00	0.00	0.00
O13p	persistent isolated class	1.00	1.00	0.00	0.00	0.94	0.00	0.00
O17	Calhoun-like run (O17)	0.99	0.83	0.92	0.00	0.72	0.00	0.00
O17p	**primary outcome**	0.99	0.83	0.00	0.00	0.70	0.00	0.00
	95% Wilson interval	0.97–1.00	0.78–0.88	0.00–0.02	0.00–0.02	0.64–0.76	0.00–0.04	0.00–0.04
O12	transplants found: phase B	52/60	46/60	47/60	–	49/60	–	–
	phase D ($1.75\,t'_{\mathrm{pk}}$)	0/60	0/60	0/60	–	0/60	–	–
	week 95, all arms	0/60	0/60	0/60	–	0/60	–	–
O12b†	D + healthy partners	0/120	0/120	0/120	–	2/120	–	–
A1–A4	anti-patterns triggered	none	none	none	none	A2: 0.02	A2: 0.55	A2: 0.03
	$\rho_{\mathrm{pk}} = N_{\mathrm{pk}}/K_\gamma$	0.36	0.33	0.42	0.71	0.42	0.77	0.85
	empty cells at peak, f_0	0.16	0.52	0.19	0.03	0.11	0.30	0.59
	persistent isolated, $\Phi_{\mathrm{BO}}^{\mathrm{pers}}$	0.17	0.16	0.01	0.09	0.13	0.00	0.00
	median T_{ext} (weeks, KM)	169	182	166	155	218	–	186
	spread of T_{ext} (KM IQR/median)	0.09	0.12	0.08	0.05	0.14	–	0.16
O8	senescent decline	0.62	0.21	0.82	0.97	0.00	0.09	0.37
O8b	slow initial decline	1.00	1.00	1.00	1.00	1.00	1.00	1.00
O9	old, female-biased remnant	0.45	0.29	0.44	0.49	0.30	0.21	0.37
O10b	functional growth phase	0.02	0.01	0.18	1.00	0.00	1.00	0.19
O11b	few late-born females conceive	1.00	1.00	1.00	1.00	1.00	1.00	0.97
O13b	isolated class later-born	0.00	0.00	0.00	0.00	0.00	0.00	0.00
O14	aggregation with free space	1.00	1.00	1.00	0.00	1.00	1.00	1.00
O15	infant mortality $\to 1$	0.01	0.00	0.00	0.01	0.01	0.00	0.01
O15b	conceptions without viable young	0.57	0.70	0.46	0.28	0.45	0.15	0.09
O13b	PASS or MARGINAL	0.01	0.04	0.04	0.00	1.00	0.16	0.63
	t'_{pk} (weeks, median)	34	35	33	31	62	31	45
	$t_{\mathrm{pk}} = \arg\max N$ (median)	55	61	53	44	80	35	66
	plateau length (median)	51	50	54	59	28	9	45
	t_{B99} (median)	40	44	38	33	86	980	55
	95% CI of the KM median	167–171	180–185	165–167	154–156	217–222	–	181–190
	$P_{\mathrm{ext}}(T)$	1.00	0.99	1.00	1.00	0.98	0.47	0.98
	CV of T_{ext} (extinct runs)	0.07	0.16	0.07	0.06	0.12	0.45	0.20
	$(T_{\mathrm{ext}} - t_{\mathrm{pk}})/t_{\mathrm{pk}}$ (extinct)	2.13	2.15	2.16	2.60	1.88	2.78	2.02

old at $1.75\,t'_{\mathrm{pk}}$ (week 60), 59–80 at $1.75\,t_{\mathrm{pk}}$ (week 97) and 65–80 at week 95, with the fertile window closing at 80 weeks; in arm X the founding rates are 51/60, 12/60 and 5/60 (McNemar $p = 4 \times 10^{-16}$, 2×10^{-4} and 0.03). Q4: the phase-D law explains why O8 depends on longevity, since O8 passes in 0.15, 0.30, 0.67, 0.83 and 0.92 of the runs for $a_{\max} = 90$, 105, 120, 135 and 150 while every essential pattern is unchanged (O17p 0.93–1.00). Q5: O13 fails only for the strongest aversion with the strongest attraction ($\kappa = 0.3$, $\alpha = 8$; $n^* \approx 3.2$), where the crowded fraction falls to the 10% threshold. Q6: profile-likelihood intervals of θ are $[-0.15, -0.10]$ binomial and $[-0.20, -0.05]$ quasi-binomial ($\phi = 3.1$) with the primary evaluator, and $[0.20, 0.40]$ and $[0.15, 0.50]$ ($\phi = 1.8$) with the preregistered evaluator; the additive model in standardized $\log r$ and $\log a_w$ with quadratic terms beats the single-index model by $\Delta\mathrm{QAIC} = 27.8$ with the primary evaluator, whereas with the preregistered evaluator the single-index model is preferred by $\Delta\mathrm{QAIC} = 11$. Q7: the logistic fit of the preregistered plane places the extinction boundary at $\gamma = 0$ at $w + 0.24\,\Lambda = 0.215$ ($\hat{\sigma} = 4.65$, $\hat{\varphi} = 1.11$ in the form $\sigma w + \varphi\Lambda = 1$); in the subcritical corner the median peak is 6 500–11 500 individuals at weeks 77–96 and O4 fails in every run. With the window varied ($a_w \in \{6, 12, 24\}$, 54

TABLE S4. Discriminating power of all criteria, essential (top) and desirable (bottom); primary evaluator, fraction of all runs that pass, NA counting as failure. Columns: the reference ensemble (C1, C1′, $\alpha = 0$, C0), the four null models with the initial condition of C1 ($\gamma = 0$; −R2*: $w = 1$ without social learning; long-lived control; no rules: C1 without its four rules), $n = 200$ each, and the short-lived control ($n = 100$). "Generic": passes in at least two of the four null models. "Discriminates": upper Wilson bound below 0.8 and Newcombe interval of the difference with C1 excluding zero (long: long-lived control; none: no rules; short: short-lived control). O11 is scored at six weeks of age.

	C1	C1′	γ=0	α=0	C0	−R2*	long	none	short	generic	discriminates against
O4	1.00	1.00	0.00	0.00	1.00	0.99	0.17	1.00	0.87	yes	γ=0, α=0, long
O5	0.99	0.85	1.00	1.00	1.00	0.97	0.89	0.91	0.25	yes	short
O6	1.00	1.00	0.00	1.00	1.00	1.00	0.99	1.00	0.45	yes	γ=0, short
O7	1.00	0.99	0.00	1.00	1.00	1.00	0.99	1.00	0.47	yes	γ=0, short
O10	1.00	1.00	0.00	1.00	1.00	1.00	1.00	1.00	1.00	yes	γ=0
O11	1.00	0.97	0.00	1.00	1.00	0.00	0.00	0.00	0.00	no	γ=0, −R2, long, none, short
O13	1.00	1.00	0.00	1.00	0.92	1.00	0.00	0.00	0.00	no	γ=0, long, none, short
O13p	1.00	1.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	no	γ=0, α=0, C0, long, none, short
O17p	0.99	0.83	0.00	0.00	0.00	0.00	0.00	0.00	0.00	no	γ=0, α=0, C0, −R2, long, none, short
O10b	0.02	0.01	1.00	1.00	0.18	0.05	0.20	0.12	1.00	no	none
O8	0.62	0.21	0.00	0.97	0.82	0.69	0.39	0.64	0.04	no	C1′, γ=0, long, short
O8b	1.00	1.00	0.00	1.00	1.00	1.00	1.00	1.00	1.00	yes	γ=0
O9	0.45	0.29	0.00	0.49	0.44	0.51	0.39	0.45	0.18	no	C1′, γ=0, short
O11b	1.00	1.00	1.00	1.00	1.00	0.99	0.95	0.93	1.00	yes	none
O14	1.00	1.00	1.00	0.00	1.00	1.00	1.00	1.00	1.00	yes	α=0

TABLE S5. Discriminating power of the redefined criteria with the preregistered evaluator (same ensembles and columns as Table S4). The preregistered O10 passes in every model; the preregistered O11 passes vacuously without crowding damage.

	C1	C1′	γ=0	α=0	C0	−R2*	long	none	short	generic	discriminates against
O4	1.00	1.00	0.00	1.00	1.00	1.00	0.17	1.00	0.87	yes	γ=0, long
O5	1.00	0.99	0.00	1.00	1.00	1.00	0.96	0.99	0.16	yes	γ=0, short
O10	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	yes	none
O11	0.98	0.94	1.00	1.00	0.98	0.15	0.09	0.21	0.00	no	−R2, long, none, short
O13p	1.00	1.00	0.00	0.01	0.00	1.00	0.00	0.00	0.00	no	γ=0, α=0, C0, long, none, short
O17p	0.98	0.93	0.00	0.01	0.00	0.15	0.00	0.00	0.00	no	γ=0, α=0, C0, −R2, long, none, short

points, 20 runs each) 51 of the 54 points have $P_{\text{ext}} \in \{0, 1\}$, so the logistic model in $\log r + \theta \log a_w$ is separated and its profile interval ($\hat{\theta} = 0.725$ [0.70, 0.75]) reflects the geometry of the grid, not sampling error; the non-parametric brackets of $\Lambda_{1/2}$ coincide for $a_w = 12$ and 24 and lie one grid step lower for $a_w = 6$ at every inheritance weight (interpolated local exponents 0.51–0.70 between 6 and 12 weeks and 0.88–1.00 between 12 and 24).

S5. TRANSPLANT EXPERIMENTS

Each of the 60 replicates per arm runs a mother colony and samples it at several times (Table S8): in phase B; at $1.75\,t'_{\text{pk}}$ (phase D of the primary evaluator); at $1.75\,t_{\text{pk}}$ with $t_{\text{pk}} = \arg\max N$ (the preregistered protocol); at week 95 in every arm, which removes the confound between rule and sampling time; and at the first week in which four males and four females past the plasticity window have $0.2 \le s \le 0.5$ (the "band"). Four males and four females are moved to an empty lattice; success is reaching 80 individuals within 150 weeks. In the mixed phases the animals of one sex are paired with healthy animals of the other. Arm X shares its mother colonies and its transplanted animals with C1 and places them in an enclosure without R1 and R2 with $r = 0.1$. Whenever R1 is present, phase-D animals never found a colony at any of the three sampling times, nor with healthy partners (0/120 at week 95 in C1, C1′, C0 and −R2; 3/120 in C2); their social state is 0 (median), with all eight below 10^{-3} in 55–93% of the trials. Without R1, at week 60 the same test gives 0/60 too, because the colonies are still at the peak and the animals are young but fully deteriorated; at $1.75\,t_{\text{pk}}$ (weeks 126–131) 10/60 recover alone in −R1 and 3/60 in −R1−R2, and with healthy partners 34/120 and 14/120 at week 95. Band animals (weeks 8–11, ages 12–15, $s = 0.33$–0.40) found colonies in 5/60 (C1), 2/60 (C1′), 0/60 (C0), 7/60 (−R2) and 8/60 (C2) under R1, against 50/60 without R1, 46/60 without R1 and R2 and 57/60 in arm X; in the colonies they found, the adult descendants have a mean social state of 0.28–0.45 in C1, 0.26–0.42 in C1′, 0.11–0.35 in C0 and 0.33–0.37 in arms X and −R1. Over 12 points of the plane (α, κ) with

TABLE S6. O17p by arm and evaluator (fraction of runs). Primary: t'_{pk}, t_{B99}, O4 with free space, $m > 8$, O10 on adults, O11 on the second half of the births. Prereg.: the full preregistered evaluator. The other columns change one option of the primary evaluator: peak time at $\arg\max N$ or at the end of the plateau; crowded class at $m > m_c$; O5 on the last sustained birth; the preregistered O4.

Arm	primary	prereg.	arg max	plateau end	$m > m_c$	O5 on t_{last}	O4 prereg.
C1	0.99	0.98	1.00	1.00	0.99	0.48	0.99
C1′	0.83	0.93	0.96	0.96	0.83	0.36	0.83
C2	0.70	0.88	0.73	0.81	0.70	0.23	0.78
C1 − AV	0.79	0.90	0.96	0.97	0.79	0.44	0.79
C1 − R1	0.21	0.23	0.22	0.22	0.21	0.12	0.21
C1 − R2	0.00	0.17	0.00	0.00	0.00	0.00	0.00
C1 − R7	0.00	0.00	0.00	0.00	0.00	0.00	0.00
C0	0.00	0.00	0.00	0.00	0.00	0.00	0.00
C1, $\alpha = 0$	0.00	0.01	0.00	0.00	0.00	0.00	0.00
$\gamma = 0$	0.00	0.00	0.00	0.00	0.00	0.00	0.00
−R2* ($w = 1$)	0.00	0.15	0.00	0.00	0.00	0.00	0.00
long-lived	0.00	0.00	0.00	0.00	0.00	0.00	0.00
no rules	0.00	0.00	0.00	0.00	0.00	0.00	0.00
C1, single-pass R7	0.99	0.97	1.00	1.00	0.99	0.51	0.99
C1, $\mu_{\mathrm{bg}} = 0.003$	0.55	0.45	0.88	0.99	0.55	0.04	0.55
C1, $\mu_{\mathrm{bg}} = 0.008$	0.06	0.02	0.32	0.72	0.06	0.00	0.06
C1, nest 3 wk	0.83	0.87	0.90	0.90	0.83	0.47	0.83
C1, asynchronous	1.00	1.00	1.00	1.00	1.00	0.71	1.00
C1, async., soft cap.	0.99	0.99	0.99	0.99	0.99	0.72	0.99
C1, $s_{\mathrm{base}} = 0.25$	0.08	0.64	0.07	0.08	0.08	0.04	0.08
C1, $s_{\mathrm{base}} = 0.5$	0.00	0.36	0.00	0.00	0.00	0.00	0.00
C1, ages U{4..52}	0.93	0.90	1.00	1.00	0.93	0.36	0.93
C1, ages U{4..80}	0.63	0.56	0.97	1.00	0.63	0.10	0.63
C1′, $\mu_{\mathrm{bg}} = 0.003$	0.28	0.28	0.68	0.89	0.28	0.02	0.28
C1′, nest 3 wk	0.60	0.63	0.69	0.69	0.60	0.34	0.60
C1′, asynchronous	0.94	0.98	0.99	0.99	0.94	0.39	0.94
C1′, $s_{\mathrm{base}} = 0.25$	0.26	0.53	0.30	0.30	0.26	0.17	0.26
C1′, ages U{4..52}	0.68	0.70	0.97	0.99	0.68	0.23	0.68
C1′, ages U{4..80}	0.31	0.30	0.77	0.94	0.31	0.03	0.31

40 transplants per point and phase, no phase-D transplant founded a colony (0/480), nor did any mixed transplant (0/960); phase-B transplants succeed in 0.63–1.00 of the trials, less often with stronger attraction. This reproduces Marsden's observation reported by Calhoun [1, 3], only with R1 and by construction.

S6. ROBUSTNESS VARIANTS AND SYSTEM SIZE

Table S9 lists the robustness variants and Table S10 the lattice sizes at fixed density. In every variant that loses the primary outcome the loss comes from a single criterion: O5 for the background mortality and the non-synchronous initial conditions, O11 for the nest period and for pups born above the deterioration threshold. O5 is lost through its timing clause $(t_{B99} - t'_{\mathrm{pk}})/t'_{\mathrm{pk}} \le 0.5$, not through its level clause $N(t_{B99}) \ge 0.7\,N_{\mathrm{pk}}$: at $\mu_{\mathrm{bg}} = 0.003$ the O5 verdicts of C1 are 110 PASS, 83 MARGINAL and 7 FAIL (median t'_{pk} 29 weeks against 34 in the reference ensemble, median t_{B99} 43 against 40); at 0.008, 12, 118 and 70 (t'_{pk} 26, t_{B99} 50); with ages U{4..80}, 126, 73 and 1 (t'_{pk} 29, t_{B99} 41); with U{4..52}, 186, 14 and 0. The column "disc." shows that the criteria that discriminate against the null models (with O11 scored at six weeks) are untouched by the demographic variants (1.00 in every C1 arm with background mortality or spread ages; 0.96–0.99 in C1′) and are lost only where O11 is lost. Along the background-mortality curve the primary outcome is lost gradually: at $\mu_{\mathrm{bg}} = 0.001$ per week, a tenth of the senescent hazard, O17p is still 0.83. In the lattice-size stage the failures at $L = 15$ (11 of 60) and $L = 20$ (6 of 60) are O5 marginal in 5 runs at each size, O11 marginal in 1 at each, and at $L = 15$ the anti-pattern A2 (recovery of natality) in 6 runs; at $L = 30$ one run fails O5 and at $L \ge 45$ none fails.

a. Global sensitivity. Fig. S1 gives the Sobol' total and first-order indices of O17p over the 11 factors of the Saltelli design (the 8 retained by the Morris screening plus w, r and a_w; $N = 256$, 6 runs per point, 19 968 runs), with bootstrap intervals and the empirical noise floor, and the Calhoun-like fraction of the two Latin hypercubes in

TABLE S7. The confirmatory family with the primary evaluator (left) and with the preregistered evaluator (right), on the same runs. Holm correction over seven tests within each column set; the p of Q6 is quasi-binomial. The preregistered evaluator takes the transplant of Q1 at $1.75\,t_{\mathrm{pk}}$ with $t_{\mathrm{pk}} = \arg\max N$, as preregistered. Q1: structural irreversibility (paired cross-transplant, exact McNemar); Q2: each rule necessary (intersection–union test over the single ablations); Q3: near-deterministic collapse against the short-lived control; Q4: phase-D law, slope of $T_{\mathrm{ext}} - t_{\mathrm{last}}$ on $a_{\max}$ (TOST, margins $[0.8, 1.25]$); Q5: O13 band centred at $n^* \approx m_c$ (Firth regression, $b_2 < 0$); Q6: a_w weighs less than r (profile likelihood, $0 < \theta < 1$); Q7: endogenous collapse at $\gamma = 0$ in the subcritical corner.

	primary evaluator				preregistered evaluator			
	Estimate	p_{Holm}	Rej.	Met	Estimate	p_{Holm}	Rej.	Met
Q1	0/60 vs. 51/60 (P_T = 0.85 [0.74, 0.92]); discordant pairs 51:0	1.3×10^{-15}	yes	yes	0/60 vs. 12/60 (P_T = 0.20 [0.12, 0.32]); discordant pairs 12:0	7.3×10^{-4}	yes	no
Q2	0.99 vs. −R1 0.21, −R2 0.00, −AV 0.79, −R7 0.00	2.2×10^{-7}	yes	no	0.97 vs. −R1 0.23, −R2 0.17, −AV 0.90, −R7 0.00	0.067	no	no
Q3	200/200 vs. 47/100; CV 0.066 [0.059, 0.071]	$< 10^{-16}$	yes	yes	200/200 vs. 47/100; CV 0.066 [0.059, 0.071]	$< 10^{-16}$	yes	yes
Q4	slope 0.996 ± 0.014; $\bar{r} = -2.0 \pm 0.3$ wk	$< 10^{-16}$	yes	yes	slope 0.996 ± 0.014; $\bar{r} = -2.0 \pm 0.3$ wk	$< 10^{-16}$	yes	yes
Q5	$b_2 = +185.3 \pm 64.3$ (convex); no interior maximum	1.00	no	no	$b_2 = +69.5 \pm 14.2$ (convex); no interior maximum	1.00	no	no
Q6	$\hat{\theta} = -0.15$ [-0.20, -0.05] (quasi-binomial, $\phi = 3.1$)	$< 10^{-16}$	yes	no	$\hat{\theta} = 0.30$ [0.15, 0.50] (quasi-binomial, $\phi = 1.8$)	4.0×10^{-5}	yes	yes
Q7	120/120 vs. 0/30 (30/30 explode)	$< 10^{-16}$	yes	yes	120/120 vs. 0/30 (30/30 explode)	$< 10^{-16}$	yes	yes

TABLE S8. Transplants that found a colony, by arm and phase (successes/trials; 60 replicates per arm). Phase D is sampled at $1.75\,t'_{\mathrm{pk}}$ (median sampling week in parentheses), then at $1.75\,t_{\mathrm{pk}}$ with $t_{\mathrm{pk}} = \arg\max N$, and at week 95 in every arm; "+healthy (95)" pools the two mixed phases at week 95 (h.: healthy partners of the other sex). The band samples eight adults past the plasticity window with $0.2 \le s \le 0.5$ at the first week in which they exist (median week in parentheses), alone or with healthy partners of the other sex.

Arm	B	D	$1.75\,t_{\mathrm{pk}}$	week 95	+healthy (95)	band	band M + h. F	band F + h. M
C1	52/60	0/60 (60)	0/60	0/60	0/120	5/60 (9)	37/60	17/60
C1′	46/60	0/60 (63)	0/60	0/60	0/120	2/60 (8)	35/60	11/60
C1 − R1	58/60	0/60 (86)	10/60	0/60	34/120	50/60 (9)	57/60	56/60
C1 − R2	50/60	0/60 (65)	0/60	0/60	0/120	7/60 (10)	39/60	7/60
C1 − R1 − R2	58/60	0/60 (82)	3/60	1/60	14/120	46/60 (9)	54/60	43/60
X (C1 animals)	59/60	51/60 (60)	12/60	5/60	41/120	57/60 (9)	60/60	58/60
C0	47/60	0/60 (58)	0/60	0/60	0/120	0/60 (11)	39/60	18/60
C2	49/60	0/60 (110)	0/60	0/60	3/120	8/60 (10)	46/60	32/60

the nested subsets chosen after the runs. The total indices of O17p are c_R 0.49 ± 0.16, w 0.43 ± 0.15, κ 0.41 ± 0.14, r 0.32 ± 0.12 and m_c 0.31 ± 0.12, followed by η, γ, p, p_0 and α (0.12–0.17, against floors of 0.05–0.09); a_w (0.11) does not exceed its floor. The first-order indices add up to 0.15 and the total indices to 2.8, and 31% of the run-level variance is within points. For the continuous outputs the first-order indices add up to 1.01 (ρ_{pk}), 0.76 (f_0) and 0.89 (persistent isolated fraction). The Morris screening ranked c_R ($\mu^* = 0.18$), κ (0.16), m_c (0.15), p_0 (0.09) and α (0.08) first for O17p. In the hypercubes, a Firth logistic regression of O17p on the standardized factors (quasi-binomial correction, false discovery rate 10%) finds the effects of κ (−), f_R (+), π (+), w (−), m_c (−) and $a_{\max}$ (+) in both hypercubes. The full decomposition is in the extended version in the repository.

S7. ENSEMBLES

Table S11 lists every ensemble of the production program with its size, horizon and base seed, so that each number of the main text can be traced to its source. Seeds are derived as `SeedSequence([base_seed, point, replicate])`; the base seeds of the production runs (25301–25336) differ from those of the runs of the preregistered design, so the two sets of runs are independent. The program comprises 51 828 simulations and 960 transplant replicates in 26 stages, about 4.4 hours of logged wall time on four cores, 1.8 of them for the Sobol' design; a run of `C1` or one of its variants costs 0.5–2.6 s of CPU time.

TABLE S9. Robustness variants ($n = 200$ per arm). O17p with 95% Wilson interval and Newcombe interval of the difference with the reference ensemble; the same arm with the preregistered evaluator and with $\arg\max N$ as peak time; "disc.": O17p restricted to the criteria that discriminate against the null models (O11 scored at six weeks, O13 and O13p pass and no anti-pattern is triggered), which leaves out the generic O4–O7 and O10; the criterion with the lowest pass rate when O17p < 0.95 (NA counted as failure); median peak density, plateau length (weeks within 2% of the maximum) and Kaplan–Meier median of T_{ext}.

Variant	O17p	Δ vs. reference	prereg.	arg max	disc.	fails	ρ_{pk}	plateau	T_{ext}
C1, reference	0.99 [0.97, 1.00]	–	0.98	1.00	1.00	–	0.36	51	169
single-pass R7 admission	0.99 [0.97, 1.00]	+0.00 [-0.02, +0.02]	0.97	1.00	1.00	–	0.37	52	168
$\mu_{\text{bg}} = 0.003$	0.55 [0.48, 0.62]	-0.44 [-0.51, -0.37]	0.45	0.88	1.00	O5 0.55	0.33	17	174
$\mu_{\text{bg}} = 0.008$	0.06 [0.03, 0.10]	-0.94 [-0.96, -0.89]	0.02	0.32	1.00	O5 0.06	0.30	10	182
nest period 3 wk	0.83 [0.78, 0.88]	-0.16 [-0.22, -0.11]	0.87	0.90	0.90	O11 0.90	0.31	53	171
asynchronous, hard	1.00 [0.98, 1.00]	+0.01 [-0.01, +0.03]	1.00	1.00	1.00	–	0.43	57	159
asynchronous, soft	0.99 [0.97, 1.00]	+0.00 [-0.02, +0.02]	0.99	0.99	0.99	–	0.42	57	158
$s_{\text{base}} = 0.25$	0.08 [0.05, 0.13]	-0.92 [-0.95, -0.86]	0.64	0.07	0.08	O11 0.08	0.40	53	164
$s_{\text{base}} = 0.5$	0.00 [0.00, 0.02]	-0.99 [-1.00, -0.97]	0.36	0.00	0.00	O11 0.00	0.44	53	165
ages U{4..52}	0.93 [0.89, 0.96]	-0.06 [-0.11, -0.03]	0.90	1.00	1.00	O5 0.93	0.36	28	167
ages U{4..80}	0.63 [0.56, 0.69]	-0.36 [-0.43, -0.30]	0.56	0.97	1.00	O5 0.63	0.34	22	173
C1′, reference	0.83 [0.78, 0.88]	–	0.93	0.96	0.97	–	0.32	50	182
single-pass R7 admission	0.85 [0.79, 0.89]	+0.02 [-0.06, +0.09]	0.91	0.95	0.96	O5 0.89	0.33	49	180
$\mu_{\text{bg}} = 0.003$	0.28 [0.22, 0.34]	-0.56 [-0.63, -0.47]	0.28	0.68	0.96	O5 0.28	0.29	19	183
$\mu_{\text{bg}} = 0.008$	0.04 [0.02, 0.08]	-0.79 [-0.84, -0.73]	0.01	0.24	0.96	O5 0.04	0.26	9	182
nest period 3 wk	0.60 [0.54, 0.67]	-0.23 [-0.31, -0.14]	0.63	0.69	0.69	O11 0.69	0.30	52	181
asynchronous, hard	0.94 [0.89, 0.96]	+0.10 [+0.04, +0.16]	0.98	0.99	0.99	O5 0.94	0.38	56	173
asynchronous, soft	0.97 [0.94, 0.99]	+0.14 [+0.08, +0.19]	0.98	1.00	1.00	–	0.38	56	173
$s_{\text{base}} = 0.25$	0.26 [0.20, 0.32]	-0.57 [-0.65, -0.49]	0.53	0.30	0.31	O11 0.32	0.38	50	175
$s_{\text{base}} = 0.5$	0.00 [0.00, 0.02]	-0.83 [-0.88, -0.77]	0.38	0.01	0.00	O11 0.01	0.43	47	174
ages U{4..52}	0.68 [0.61, 0.74]	-0.15 [-0.24, -0.07]	0.70	0.97	0.99	O5 0.69	0.33	26	182
ages U{4..80}	0.31 [0.25, 0.38]	-0.52 [-0.60, -0.44]	0.30	0.77	0.99	O5 0.31	0.30	21	187

TABLE S10. Lattice side L at fixed seeding density $N_0/L^2 = 0.44$ ($n = 60$ per size, $T = 600$). Medians over runs; t_{last} is the last surviving birth, $D_{\text{last}} = T_{\text{ext}} - t_{\text{last}}$; the spread is the interquartile range over the median of the Kaplan–Meier estimate of T_{ext}, with bootstrap interval.

L	N_0	n	O17p	ρ_{pk}	f_0	$\Phi_{\text{BO}}^{\text{pers}}$	t_{last}	IQR	D_{last}	T_{ext}	spread
15	100	60	0.82 [0.70, 0.89]	0.35	0.18	0.18	50	29	110	162	0.148 [0.121, 0.194]
20	178	60	0.90 [0.80, 0.95]	0.36	0.17	0.17	62	22	112	175	0.143 [0.110, 0.169]
30	400	60	0.98 [0.91, 1.00]	0.37	0.16	0.17	58	13	110	170	0.076 [0.053, 0.112]
45	900	60	1.00 [0.94, 1.00]	0.37	0.15	0.17	66	16	111	176	0.085 [0.052, 0.113]
60	1600	60	1.00 [0.94, 1.00]	0.37	0.15	0.17	75	14	110	184	0.076 [0.050, 0.103]

S8. DEVIATIONS FROM THE PREREGISTRATION

Table S12 lists every deviation from the preregistered plan (`docs/preregistered_plan.md` in the code repository, dated 1 October 2026, 22:02 UTC, before the launch of the production runs), including the redefinition of the primary evaluator, and the analyses labelled post hoc.

S9. CONDENSED MODEL DESCRIPTION (ODD PROTOCOL)

We follow the second update of the ODD protocol [4]; the full description is in the extended version in the repository. Parameter names refer to the `Params` class of the `u25` package.

a. Purpose and patterns. The purpose is to identify a minimal set of *local* rules that, with unlimited resources and without thresholds on the total population, generates the qualitative sequence reported for Universe 25: latency, exponential growth, deceleration with social deterioration, cessation of surviving births near the peak, decline without recovery to extinction, and differentiated classes of deteriorated animals. The model is evaluated against the patterns of Table S1 and, for each pattern, against null models. With the default initial condition only items 3–6 are testable, and the model reproduces them as a boom of one or two generations followed by extinction by senescence; of these items, the crowding damage and the demography already give the end of natality and the decline, and the rules add the classes.

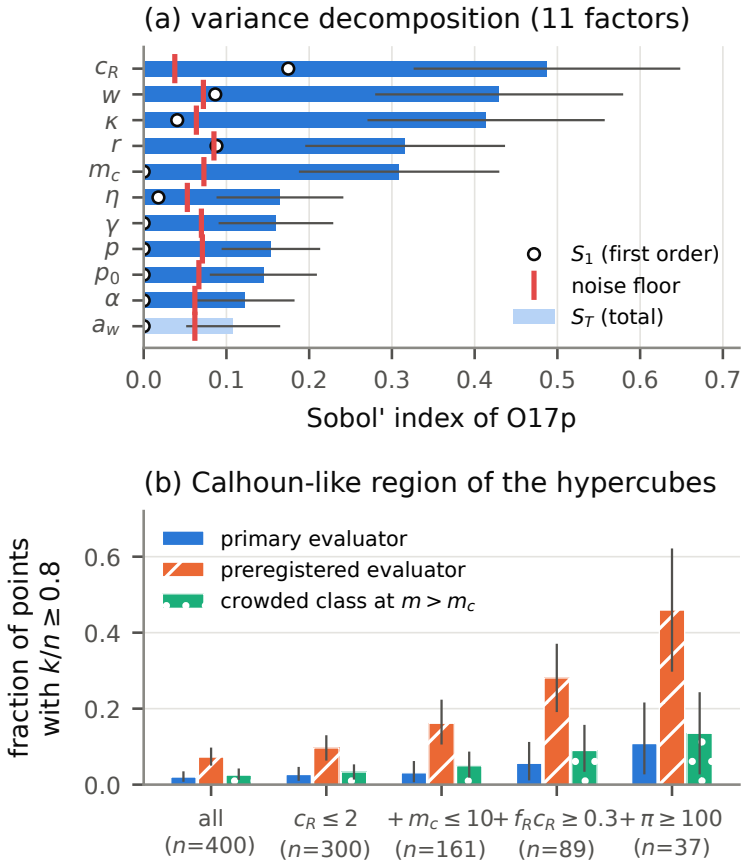


FIG. S1. Global sensitivity around C1 (primary evaluator). **(a)** Sobol' total indices S_T of O17p (bars, bootstrap intervals), first-order indices S_1 (open circles) and empirical noise floor (red ticks); Saltelli design with $N = 256$, 11 factors, 6 runs per point. Dark bars meet the relevance criterion $S_T - S_T^{\text{noise}} > 0.05$ with the lower confidence bound above the floor. **(b)** Fraction of Calhoun-like points ($k/n \geq 0.8$) of the two Latin hypercubes in nested subsets chosen after the runs ($c_R \leq 2$, then $m_c \leq 10$, $f_R c_R \geq 0.3$ and $\pi \geq 100$), with bootstrap intervals, for the primary evaluator, the preregistered evaluator and the primary evaluator with the crowded class at $m > m_c$.

b. Entities, state variables and scales. Each agent carries an identifier, a sex $\sigma_i \in \{M, F\}$ fixed at birth, a cell (x_i, y_i) on an $L \times L$ lattice with hard walls ($L = 30$), an age a_i in weeks, a social state $s_i \in [0, 1]$ (1 = full social competence) and, in C2, the maternal social state at birth h_i. Each pair carries a symmetric affinity $F_{ij} \in [0, 1]$, stored sparsely (pairs with $F_{ij} < \varepsilon$ are pruned). Ordinary cells have no occupancy limit; a fixed random fraction f_R of cells are *refuges* with capacity c_R, drawn once (`refuge_seed` $= 0$) and identical in every run (eight alternative layouts gave no detectable difference). One time step is one week. A cell is an abstract meeting place, mapped to a nest box of Universe 25, the damage threshold m_c to its comfortable occupancy and $K_\gamma = L^2 m_c = 7200$, the population at which every cell would sit at the damage threshold, to Calhoun's 3840 places; K_γ is a reference for the peak density, not a physical capacity, and $|G|$ and m_c were not calibrated. The refuges correspond to the little-preferred high nests; their capacity $c_R = 2$ is the isolation threshold of O13, not the capacity of a nest. Mobility is at most one Moore step per week, so the ratio of the mixing time to the deterioration time (a Damköhler number) is about 1.7×10^2 in C1, against $\lesssim 4 \times 10^{-3}$ in Universe 25.

c. Process overview and scheduling. Each week, in order: (1) movement, simultaneous, with weights evaluated on the pre-move snapshot, with refuge capacity enforced after the move (an asynchronous variant, agents moving one at a time in random order, is a robustness check); (2) affinity update; (3) occupancy m_i after movement, including i itself; (4) social-state update; (5) reproduction; (6) ageing; (7) mortality; (8) recording of observables. Updates within each submodel are synchronous.

d. Design concepts. *Basic principles:* social stress from crowding reduces reproduction and offspring survival [5, 6]; maternal transmission of individual quality [7, 8]; the behavioural sink, attraction to familiar individuals producing aggregation despite free space [1, 9]. *Emergence:* the population trajectory, the spatial aggregation, the effective deterioration threshold and the classes of crowded and isolated deteriorated animals are not imposed by any

TABLE S11. Ensembles of the production program (`results_v2/f2b_<stage>` in the code repository). Points: design points or arms; per point: runs or transplant replicates; T: horizon in weeks; seed: base seed of the stage.

Stage	Content	points	per point	runs	T	seed	used for
`base`	reference ensemble: C1, C1′, C0, $\alpha = 0$	4	200	800	600	25301	production
`c1c2`	C1 and C2	2	200	400	700	25318	QC
`ref`	short-lived and long-lived controls	2	100	200	1000	25302	Q3, controls
`nulos`	four null models	4	200	800	600	25330	discrimination
`abl1`	single ablations and variants	9	120	1080	600	25303	Q2
`abl2`	double to quadruple ablations, $\gamma = 0$	12	40	480	600	25304	ablations
`trans`	8 transplant arms, 11 phases	8	60	480	300	25306	Q1, O12
`amax`	$a_{\max} \in \{90, \dots, 150\}$	5	60	300	600	25305	Q4
`rob`	demographic and update variants	16	200	3200	700	25331	robustness
`ci`	non-synchronous initial condition	4	200	800	700	25333	robustness
`size`	$L \in \{15, \dots, 60\}$ at fixed density	5	60	300	600	25334	size
`q7aw`	a_w varied at $\gamma = 0$	54	20	1080	600	25335	Q7
`p1`	plane $(r, w) \times \gamma$	112	30	3360	600	25309	Q7, P1
`p1aw`	plane (r, a_w)	56	30	1680	600	25310	Q6
`p2`	plane (α, κ), p	72	30	2160	600	25308	Q5, P2
`p5`	refuge plane	50	30	1500	600	25311	P5
`lhs`	Latin hypercube 1	200	10	2000	400	25312	global
`lhs2`	Latin hypercube 2	200	10	2000	400	25313	global
`lhsx`	boundary points of the hypercubes	18	40	720	400	25336	global
`morris`	Morris screening	320	10	3200	400	25307	global
`sobol`	Saltelli design, 11 factors	3328	6	19968	400	25314	global
`robmu`	μ_{bg} curve	12	60	720	700	25332	robustness
`p4`	seeding density and lattice	24	30	720	600	25315	P4
`trgrid`	transplants on (α, κ)	12	40	480	300	25316	O12
`refine`	40 boundary points of the planes	40	100	4000	600	25317	planes
`p6`	founder protocol	12	30	360	900	25319	O1–O3

rule; imposed by design, and therefore consistency checks, are the irreversibility of adults (R1) and with it the failure of transplanted late-phase animals, the persistence of the isolated class in refuges of capacity $c_R = 2$, and the absence of social deaths and the senescent decline, since social mortality is off. *Adaptation, objectives, learning, prediction:* agents do not optimize anything; "social learning" (R2) is a contagion rule. *Sensing:* agents sense only their Moore neighbourhood (their affinities with the occupants of each neighbouring cell, the occupancy of those cells, the mean social state of neighbours, and whether a cell is a refuge and is full); no rule uses the total population size. *Interaction:* co-presence (affinity), mating within the cell, social learning and crowding. *Stochasticity:* destination choice; mate choice, conception, litter size, neonatal survival and sex; death; the order of entry into full refuges and the relocation cell when a refuge is full; in the asynchronous variant the order of moves; in the non-synchronous variant the initial ages. The state $N = 0$ is absorbing. *Observation:* per-step time series, per-run summaries and the automatic pattern evaluator.

e. Initialization and input. Default: 200 males and 200 females aged 4 weeks with $s = 1$, placed uniformly at random, with no affinities (a synchronous cohort; integer ages drawn uniformly between 4 and 52 or between 4 and 80 weeks are a robustness check). Founders: 4 males and 4 females aged 7 weeks with $s = 1$ in the central cell. Transplant: 4 males and 4 females of reproductive age taken from a donor run, keeping their s, placed in the central cell of an empty lattice. There is no input data: no resource limitation, predation or emigration.

f. Submodels. Movement. Agent i chooses a destination y in its Moore neighbourhood $V(i)$, which includes its own cell, with weight

$$w_i(y) = \big[1 + \alpha A_i(y)\big]\, e^{-\kappa(1-s_i)^p n_y}\, \big[1 + \pi\,(1-b_i)^q\, \mathbf{1}_{\mathcal{R}}(y)\big], \qquad A_i(y) = \sum_{j \in y,\, j \neq i} F_{ij},$$

where n_y is the occupancy of y excluding i, $\mathbf{1}_{\mathcal{R}}(y) = 1$ if y is a non-full refuge ($n_y < c_R$) and 0 otherwise, κ and p define the aversion of the deteriorated (AV), π and q the refuge preference (R7), and $b_i = s_i$ in C1 and $b_i = h_i$ in C2. Full refuges cannot be entered: residents have priority, entering agents are drawn in random order, rejected agents return to their previous cell, and admission is iterated to a fixed point (`refuge_hard="strict"`); pups, founders and transplanted animals that would put a refuge above capacity are placed in a uniformly chosen non-refuge cell of the nearest Chebyshev ring that contains one. A single-pass admission, which does not count returning agents and newborns, is kept as a robustness variant. *Affinity.* If i and j are at Chebyshev distance ≤ 1 after movement, $F_{ij} \leftarrow (1-\eta)F_{ij} + \eta$; otherwise $F_{ij} \leftarrow (1-\delta)F_{ij}$. *Social state.*

$$s_i \leftarrow \Pi_{[0,1]}\Big(s_i + r\,\mathbf{1}_{a_i \leq a_w}\big[(1-\lambda) + \lambda\,\bar{s}_{V(i)}\big] - \gamma\,(m_i - m_c)_+\Big),$$

where r is the recovery (learning) rate, $\bar{s}_{V(i)}$ the mean social state of the *other* agents in $V(i)$ (zero if there are none), a_w the plasticity window (R1) and λ the social-learning weight (R2); the neutral values $a_w = \infty$ and $\lambda = 0$ (with

$w = 1$ and $r = 0.01$) recover the core model. *Reproduction.* Each reproductive female ($a_{\min} \le a \le a_{\text{rep}}$) that shares her cell with at least one reproductive male chooses one uniformly at random and conceives with probability $p_0 s_f s_m$; the litter size is drawn from an empirical distribution on $\{3, \dots, 10\}$ (mean 6.55) and Binomial($L, q_0 s_f$) pups survive, with $q_0 = 0.95$. Pups are born in the mother's cell at age 0, with sex ratio 1/2 and social state $s = w\, s_f + (1-w)\, s_{\text{base}}$ (with $w = 0.2$ and $s_{\text{base}} = 0$ every pup is born with $s \le 0.2$); they are independent agents from birth and add to local crowding (a nest period of three weeks is a robustness check). *Mortality.* After ageing, each agent dies with probability $1/(1 + e^{-k(a-a_{\max})})$. Social mortality is switched off in all presets; a background hazard μ_{bg} at all ages, combined as $1 - (1 - p_{\text{age}})(1 - \mu_{\text{bg}})$, is a robustness check.

S10. FURTHER DETAILS OF THE MODEL

This section complements the model section of the main text and the ODD description above with the details that the main text leaves out: the refuge capacity and the update scheme, the dimensionless groups and time scales, and the candidates and rules that are not used in the main text.

A. Hard capacity of the refuges and update scheme

The admission rule of the refuges, iterated to a fixed point, is given in the ODD description above. A single-pass admission that does not count returning animals and newborns, so that a refuge can hold up to 11 individuals, is a robustness variant; it leaves the primary outcome unchanged (0.995 with either rule).

Both the preference and the hard capacity are necessary with synchronous moves. Without a preference ($\pi = 0$) the refuges stay empty, because a deteriorated individual surrounded by acquaintances prefers an occupied neighboring cell; with a soft capacity, simultaneous moves produce stampedes of deteriorated animals into the same refuge. With a random sequential update a soft capacity also produces a persistent isolated class (O17p 1.00 and 0.995 with hard and soft capacity), with a higher peak density ($\rho_{\text{pk}} = 0.43$ against 0.36). The hard capacity is therefore a remedy for the collisions of the synchronous update, not a mechanism of its own. The total refuge capacity, $f_R c_R |G| = 540$ places, is about 20% of the adults at the peak, $\phi_R = f_R c_R |G| / N_{\text{ad}}(t_{\text{pk}}) \approx 0.2$, and bounds the persistently isolated fraction, $\Phi_{\text{BO}}^{\text{pers}} \le \phi_R$. The bound holds at all 50 points of the refuge plane, so a class of $\ge 10\%$ needs $f_R c_R \gtrsim 0.3$; in that plane a persistent class appears only for $f_R c_R \ge 0.4$ and a strong preference.

B. Dimensionless groups and time scales

The values are for C1; Table S14 compares the scales with Universe 25.

- $\Pi_\gamma = \gamma / r = 1$: crowding damage per excess neighbor relative to the weekly learning rate.
- $\Lambda = r\, a_w = 1.2$: the largest competence a juvenile can learn inside the window. A socialized lineage sustains itself only if $w + \Lambda \langle \bar{s}_V \rangle \gtrsim 1$; fast growth lowers $\langle \bar{s}_V \rangle$, because novices outnumber teachers (a *learning dilution*). When a_w is varied as well, r and a_w do not act through their product alone: the local weight of a_w relative to r is about one half between $a_w = 6$ and 12 weeks and about one between 12 and 24.
- $n^*/m_c \approx 0.8$, with $n^* = 1/\kappa - 1/\alpha$ the preferred group size of a deteriorated individual. Without refuges it decides between pools, isolation, or both; with refuges it only marks the onset of homogenization, near $n^* \approx 3$–4.
- $\phi_R \approx 0.2$, the refuge capacity per adult at the peak (Sec. S10 A).
- $n_0 = N_0/|G| = 0.44$ and the Damköhler number $\text{Da} = t_{\text{mix}}/t_{\text{det}}$. Here $t_{\text{mix}} = L^2/(4D_{\text{eff}})$, with $D_{\text{eff}} \approx 0.13$ cells^2 per week the mean squared displacement over 16 weeks divided by 4×16 (8 exploratory runs), so $t_{\text{mix}} \approx 1.7 \times 10^3$ weeks; and $t_{\text{det}} = t_{1/2} - t_{\text{on}} \approx 10$ weeks is the time the animals past the window (age $\ge a_w$) need to lose half of their competence ($t_{1/2} = 16$, the first week with their mean $s < 0.5$) after crowding sets in ($t_{\text{on}} = 6$, the first week with 5% of the individuals in cells with $m > m_c$). These animals no longer learn, and newborns do not enter their mean until they leave the window, so t_{det} measures deterioration and not the dilution by pups born with $s \le 0.2$. Thus $\text{Da} \approx 1.7 \times 10^2$, and deterioration is correlated over $\xi = (4D_{\text{eff}} t_{\text{det}})^{1/2} \approx 2$ cells, the size of the contact neighborhood. In founder colonies the relevant transport is the colonization front, and $t_{\text{fill}} \approx t_{\text{pk}} \approx 170$–215 weeks against $t_{\text{det}} \approx 20$–30 gives $\text{Da}_{\text{fill}} \approx 10$. In Universe 25 mice crossed the enclosure daily and the social organization took a whole phase C, about 35 weeks, to die, so $\text{Da} \lesssim 4 \times 10^{-3}$: the spatial structure of the model emerges among animals that are almost sessile on the time scale of deterioration, which is

not Calhoun's regime. Colonies grow from local nuclei, so the peak density falls with n_0, from 0.40 at $N_0 = 400$ to 0.03 for eight founders.

- $\Gamma = t_{\rm det}/T_{\rm gen} \approx 1.1$, with $T_{\rm gen} \approx 9$ weeks the interval between the median birth weeks of the first and second generations; the interquartile range over runs is 1.0–1.29, and the founders, a closed cohort, give 1.2. The mean over all adults (age $\geq a_{\rm min}$) would give 0.8 because it is diluted by the first generation, which enters it with $s \leq 0.2$ while still inside the window. With overlapping cohorts the interval between median birth weeks is not a demographic generation time; in the null models it drops to 5 weeks, below the age of first reproduction, which inflates their Γ (1.4–1.6). With the mean age of the mothers at birth ($T_{\rm gen} = 14.3$ weeks in C1) $\Gamma \approx 0.7$, and 0.60–0.85 in the other arms where it was computed: Γ is of order one under either definition. It does not set the depth of the boom: with five times less crowding damage ($\gamma = 0.02$) $\Gamma = 1.6$ and the boom is still one to two generations deep (mean generation of the births 1.42 against 1.35, net reproductive number of the first generation 0.52, O17p in 17 of 20 runs), whereas pups born at $s_{\rm nb} \approx 0.4$ ($s_0 = 0.25$) deepen it more at about the same Γ (1.3; mean generation 1.51, net reproductive number 0.62). In Universe 25 the deterioration took the whole phase C, about 35 weeks, with $T_{\rm gen} \approx 10$ weeks, so $\Gamma \approx 3.5$ (our estimate, with a different operational definition): an order-of-magnitude comparison, not a test.

- *Extinction time.* Without mortality before senescence $T_{\rm ext} = \max_i(b_i + \ell_i) \simeq t_{\rm last} + \ell_{\rm max}$, the last birth plus the longest residual lifetime. In C1 $\ell_{\rm max} \approx 110$ weeks, with an interquartile range of a few weeks, of the order of $1/k \approx 7$ weeks, so the spread of $T_{\rm ext}$ is that of $t_{\rm last}$, set by how abruptly natality stops, $\tau_B = (t_{B99} - t_{B90})/\ln 10 \approx 7$ weeks. At fixed density the median of $T_{\rm ext}$ grows with the logarithm of the system size, from 162 weeks at $L = 15$ to 184 at $L = 60$, and its interquartile range over the median is 0.15 for $L \leq 20$ and about 0.08 for $L \geq 30$. Ratios normalized by the peak time depend on where it falls on the plateau: $(T_{\rm ext} - t_{\rm pk})/t_{\rm pk} \approx 2.1$ (Universe 25: ≥ 1.84) becomes 4.0 with the start of the plateau as peak time, and $t_{\rm pk}/a_{\rm max} \approx 0.45$ (0.28), against 0.7 in Universe 25.

C. Other candidates and rejected rules

Mortality. The full form is $p_{\rm death}(a, s) = \left[1 + e^{-k(a - a_{\rm max}[1-\mu_s(1-s)])}\right]^{-1}$, where $\mu_s \geq 0$ would advance the senescence of deteriorated individuals; every model, candidates and controls, has $\mu_s = 0$, so adult death is purely senescent. There is no gestation delay, and a female may give birth every week.

C0 is the core model with the long-lived demography and R1, R2 and AV, without refuges (`calhoun_C0`). It is the minimal model that satisfies the essential patterns evaluable with $N_0 = 400$ when O13 is read instantaneously (O17 in 0.92 of the runs), but its isolated class is transient (O17p in 0). In C0 each of R1, R2 and AV is necessary: without R2, O11 and O13 are never reproduced; without AV the deteriorated animals collapse into a few pools, 76% of the lattice is empty and the isolated class disappears; without R1 adults recover when the colony thins and O17 falls to 0.40. C2 (`calhoun_C2`) is C0 plus R7 with the retreat driven by the competence of the mother at birth, $b_i = h_i$, with $q = 1$ and $\pi = 1000$: pups of deteriorated mothers retreat, and socialized adults that later deteriorate do not. It is motivated by the beautiful ones belonging to "the last half of the population born" [1]; it makes the isolated class younger than the crowded one, but the late-cohort pattern O13b only reaches MARGINAL, and O17p falls to 0.705 [0.64, 0.76]. Basing the retreat on juvenile competence (the value of s at age a_w) fails, because under R2 almost every cohort born after the first ends the window with $s < 0.1$.

Several rules were unnecessary or destroyed some pattern, and are not part of any candidate: long-range jumps, an attraction that decreases with s (it homogenizes the population and raises the peak), territoriality and defeat of males, social mortality, bonds restricted to competent pairs, and movement sub-steps (with more sub-steps the coexistence of classes is lost).

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TABLE S12. Deviations from the preregistered plan and post hoc analyses. "Before production" means after the calibration but before the production runs; "post hoc" means after the runs had been seen. The last block lists the changes that are not preregistered but were fixed, with their decision rules, before the production runs (`docs/production_plan.md` in the code repository, 2 October 2026, 05:34 UTC).

Change	When	Affects	Treatment
O13p (persistent isolated class) defined and made part of the primary outcome, together with R7	before production, after seeing that C0 has no persistent class	O17p, Q2, necessity of R7	declared as circular for R7 (main text, Discussion)
O6: rebound count also detects slow regrowth still rising at censoring	post hoc (2 Oct., 02:44)	one C1 run of 200 under the preregistered design; none in the production runs	primary evaluator
O13b: reading of the window against the median birth week	post hoc	the interpretation of O13b in C1 and C2	labelled post hoc
Robust spread of T_{ext} (Kaplan–Meier interquartile range over median)	post hoc (the plan had CV ≤ 0.10 for Q3)	the claims about the spread of the extinction time	labelled post hoc; the equivalence of C1′ with the long-lived control found under the preregistered design does not hold in the production runs
Ensembles of C1′ and the C1–C1′ contrast	post hoc	minimality, the role of AV	labelled post hoc
Subsets of the hypercubes (nested conditions on c_R, m_c, $f_R c_R$, π)	post hoc	interpretation of the Calhoun-like fraction	descriptive only
Re-scoring with other class thresholds θ and ensemble thresholds τ (Sec. S3)	post hoc	conditionality of minimality	descriptive only
Cure model for the extinction time	post hoc	survival summaries	no test reported
Holm p-value of Q6 computed from the binomial likelihood	as planned	Q6	the quasi-binomial value enters the Holm sequence; no decision changes
Not preregistered, fixed before the production runs			
Hard refuge capacity enforced strictly (pups and rejected movers counted); the single-pass admission kept as a variant	before the production runs	every stage with refuges; all stages run with seeds different from those of the preregistered design	both admission rules reported side by side (no change in O17p)
Primary evaluator redefined: peak time at the start of the plateau (t'_{pk}); end of natality in O5 at 99% of the births; O4 also requires free space at the peak ($f_0 \geq 0.10$, an informal target of the calibration); crowded class $m > 8$, independent of m_c; O10 on the adult social state; O11 on the second half of the births, not applicable when the late cohorts are empty; NA counts as failure in O17p	fixed after exploratory runs with the primary evaluator and before the production runs	every pattern normalised by t_{pk}, O4, O5, O10, O11, O13, O13p, O17p in every stage; the verdicts of Q1 (prediction), Q2 (null) and Q6 (prediction)	the full preregistered evaluator, and each option changed separately, reported next to the primary one (Table S6)
O5 birth reference t_{B99} instead of the last sustained birth t_{last}	after exploratory runs had shown that t_{last} with t'_{pk} fails O5 in most C1 runs (0.48 in the production runs, Table S6); a choice informed by the result	Q2: it is the only option on which the rejection of its null hypothesis depends (with the preregistered evaluator 5 of 7 null hypotheses are rejected and Q2 is not; with the primary one, 6 of 7)	both birth references reported for every arm; the count with the preregistered evaluator given next to the primary one
Confirmatory test of the equivalence of the exponents of r and a_w at $\gamma = 0$ (TOST, margin $[2/3, 3/2]$, stage a_w varied)	before the production runs	Q7 with a_w varied	the design cannot carry it out as inference (51 of 54 points at $P_{\text{ext}} \in \{0, 1\}$, separated logistic); replaced by a nonparametric summary per window (Sec. S4); the planned estimates are kept, labelled, in `results_numbers.json`
O8 (senescent decline) of C1: 0.71 under the preregistered design, 0.62 in the production runs (0.64 with the single-pass admission and the seeds of the production runs)	observed after the production runs	a desirable pattern; no decision	reported; not a conclusion
Transplant of Q1 taken at $1.75\,t'_{\text{pk}}$ instead of $1.75\,t_{\text{pk}}$; sampling at a common week added	before the production runs	Q1, O12	both times reported; comparisons between arms only at equal time
Sobol' design with 11 factors (the 8 selected by Morris plus w, r, a_w)	before the production runs	global sensitivity	indices conditional on the 4 fixed factors
Additional stages: null models, demographic and update variants, nonsynchronous initial condition, transplants of intermediate animals, extra replicates at the hypercube boundary, a_w varied at $\gamma = 0$ (additional confirmatory test), lattice size	before the production runs	robustness, discrimination, Q1, Q7, feasible region	decision rules fixed before the runs

TABLE S13. Parameters of the short-lived control (SL), values in C1, and justification. [E] empirical, [O] order of magnitude, [M] modelling choice, [C] calibrated qualitatively. C1′ is C1 with $\kappa = 0$; C0 is C1 without refuges; C2 has $\pi = 1000$, $q = 1$ and the maternal basis $b_i = h_i$.

Parameter	SL	C1	Justification
`grid_size` L	30	30	[M] abstract space; Calhoun had 16 cells, 256 nest boxes
N_0	200+200	200+200	[M]; founders 4+4 as in [1]
`initial_age`	4	4	[M]; 7 for founders (48 days)
η, δ, ε	0.2, 0.01, 10^{-6}	same	[M] bond formation ∼5 weeks, memory ∼100 weeks
α	1.4	4	[C] attraction to familiar individuals (behavioural sink)
m_c	8	8	[M/O] "Fifteen mice could comfortably rest in a single nest box"; groups of "over 10 individuals" [1]
r (`rec`)	0.01	0.1	[M/C] recovery/learning rate
γ	0.1	0.1	[M] damage per excess occupant
p_0	0.3	0.3	[O] weekly conception probability given an encounter
q_0	0.95	0.95	[M]
$a_{\min}$	6	6	[O] sexual maturity ≈6 weeks
a_{rep}	50	80	[E] menopause at "about 560 days" ≈80 weeks [1]
litter sizes	3–10, mean 6.55	same	[O] mean 6.12 in the source of [10]
$a_{\max}$, k	60, 0.15	120, 0.15	[E] "large numbers" lived to 800 days; last survivors > 987 days [1]
a_w (R1)	∞	12	[C] juvenile plasticity window
λ (R2)	0	1	[C] social learning
w, s_{base}	1, 1	0.2, 0	[C] partial maternal inheritance
κ, p (AV)	0, 1	0.15, 4	[C] crowd aversion of deteriorated animals
refuge fraction, capacity	0, –	0.3, 2 (hard)	[C] R7; layout `refuge_seed` = 0
π, q, basis (R7)	0	100, 4, current	[C]; C2: 1000, 1, maternal
`max_pop`	60000	15000	numerical safeguard

TABLE S14. Scales of C1 and of Universe 25 [1]. †Our estimate. The model's t_{det} is measured on the animals past the window; for Universe 25 it is the duration of phase C, so the two values of Γ compare only in order of magnitude. Neither $|G|$ nor m_c was calibrated to Universe 25, so densities are compared only in sign.

Quantity	Model (C1)	Universe 25
Resting sites	900 cells	256 nest boxes
Comfortable occupancy	$m_c = 8$	15 per box
K_γ (sites × occupancy)	7200	3840
Isolation sites	270 refuges, 540 places	high boxes, 25–50%†
Peak; N_{pk}/K_γ	2600; 0.36	2200; 0.57
Empty sites at the peak	16%	20%
Movement	$D_{\text{eff}} \approx 0.13$ cells^2/week	enclosure crossed daily
Maturity / menopause / life	6 / 80 / 120 weeks	6–7 / 80 / 114 weeks
Generation time T_{gen}	≈ 9 weeks	≈ 10 weeks†
Deterioration time t_{det}	10 weeks	≈ 35 weeks (phase C)†
$\text{Da} = t_{\text{mix}}/t_{\text{det}}$	$\approx 1.7 \times 10^2$	$\lesssim 4 \times 10^{-3}$
$\Gamma = t_{\text{det}}/T_{\text{gen}}$	1.1	≈ 3.5†